

Week 7, Loss functions
CAPP 30254, Machine Learning for Public Policy

MACHINE LEARNING
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FOR PUBLIC POLICY FOR
SPRING 2025
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Our original machine learning goal

Recall our original linear machine learning goal:

Given the training samples $(\mathbf{x}_i, y_i)$ for $i = 1, \dots, n$, we want to learn a model that takes new data $\mathbf{x}$ and makes a class prediction $\hat{y}$.

A linear model makes predictions:

$$\hat{y} = \begin{cases} +1 & \text{if } \mathbf{w}^T \mathbf{x} \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

for a weight vector $\mathbf{w}$. How do we find a "good" $\mathbf{w}$?

Squared error loss

We want to find a weight vector $\mathbf{w}$ so that $\hat{y}_i = y_i$ for training samples i . This is accomplished by choosing a loss function ℓ and finding $\mathbf{w}$ so that ℓ is minimized.

For example, the sum of squared error (SSE) loss function ℓ is:

$$\ell(\mathbf{w}) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

function of $\underline{\mathbf{w}}$ $\hat{y}_i$ depends on $\underline{\mathbf{w}}$

$\hat{y}_i = \mathbf{w}^T \mathbf{x}_i$

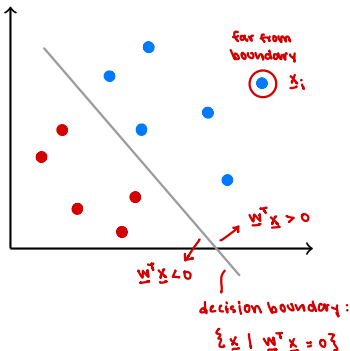
Which model uses SSE? least squares

$$\underline{\mathbf{w}}_{ls} = \underset{\underline{\mathbf{w}}}{\operatorname{argmin}} \ell(\underline{\mathbf{w}})$$

different models:
→ different loss functions
→ different results

What happens far from the boundary?

Consider training data $(\mathbf{x}_i, y_i)$ and weight vector $\mathbf{w} \in \mathbb{R}^2$. What happens to a sample far away from the boundary?



SSE for 1 sample:

$$L_i(\mathbf{w}) = (y_i - \mathbf{w}^T \mathbf{x}_i)^2$$
$$* = \left[y_i \left(1 - \frac{1}{y_i} \mathbf{w}^T \mathbf{x}_i \right) \right]^2$$

$$y_i \in \{+1, -1\}$$

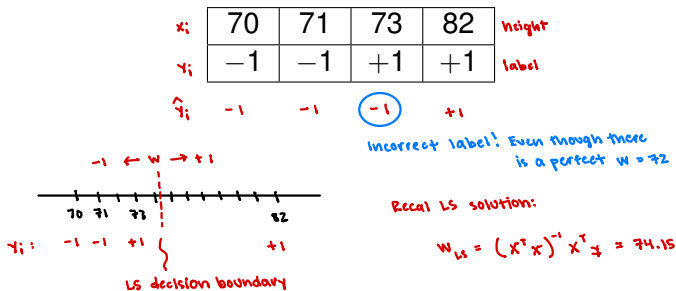
$$\text{for } \mathbf{x}_i, y_i = 1 \quad * = (1 - \mathbf{w}^T \mathbf{x}_i)^2$$
$$y_i = -1 \quad * = (1 + \mathbf{w}^T \mathbf{x}_i)^2$$

$$\text{Ex. } \mathbf{w}^T \mathbf{x}_i = 10 \quad L_i = (1 - 10)^2 > 0$$

Nonzero error even though
 $\mathbf{x}_i$ is labeled correctly!
surprise!

Given someone's height, do they play basketball?

Consider the following dataset of heights and labels: +1 for "plays basketball" and -1 for "doesn't play basketball".



The sum of squared error can do some weird things, but it's easy to work with mathematically. What other loss functions are there?

Ideal loss

Loss functions answer the question:

How well does a model make predictions?

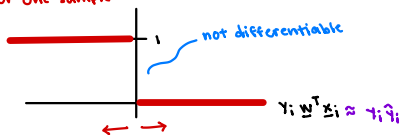
ultimate goal: $\min_{\mathbf{w}} \ell(\mathbf{w})$

The *ideal loss*, or 0/1 loss, counts the number of incorrect predictions:

$$\ell(\mathbf{w}) = \sum_{i=1}^n \mathbb{I}(y_i \neq \hat{y}_i)$$

incorrect prediction

ℓ for one sample:



indicator function

$$\mathbb{I}(z) = \begin{cases} 1 & \text{if } z \\ 0 & \text{otherwise} \end{cases}$$

prediction is incorrect, $\ell = 1$

prediction is correct
 $\ell = 0$ (no loss)

aside:

$$y_i = +1, \hat{y}_i = +1 \\ \parallel \\ \mathbf{w}^T \mathbf{x}_i > 0$$

$$\text{correct label:} \\ y_i \mathbf{w}^T \mathbf{x}_i > 0$$

$$y_i = +1, \hat{y}_i = -1 \\ \parallel \\ \mathbf{w}^T \mathbf{x}_i < 0$$

$$\text{incorrect label:} \\ y_i \mathbf{w}^T \mathbf{x}_i < 0$$

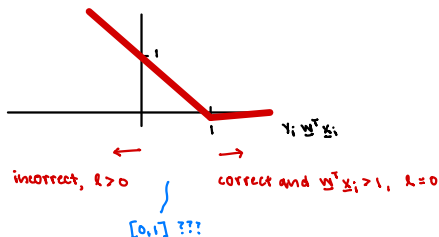
The 0/1 loss is a good measure, but it's difficult to work with mathematically.

need gradient for SGD

Hinge loss

Another common loss function, the *Hinge loss*, is given by:

$$\ell(\mathbf{w}) = \sum_{i=1}^n \max(0, 1 - y_i \cdot \hat{y}_i)$$



ℓ is not differentiable, but
we can take the "subgradient":

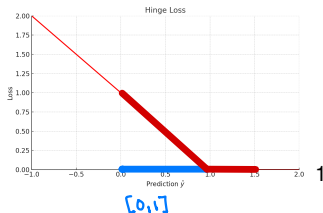
$$\nabla_{\mathbf{w}} \ell = \sum_{i=1}^n \mathbb{I}(y_i \cdot \hat{y}_i < 1) (-y_i \cdot \mathbf{x}_i)$$

Hinge mimics 0/1 loss, but is convex and easy to minimize.

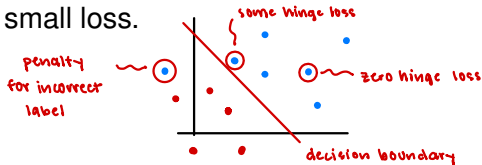
This is close to the ideal loss *and* we can compute minimums.

Hinge loss between $[0, 1]$

What happens between $[0, 1]$ of the hinge loss?

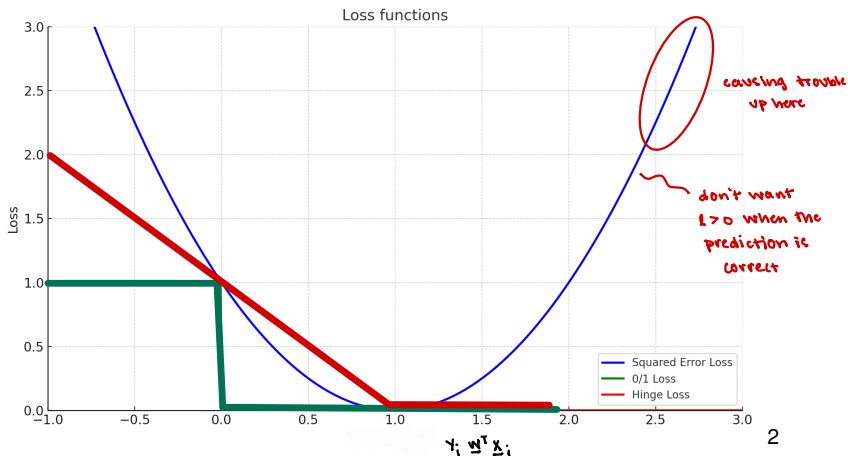


In this range, the prediction was correct, but the prediction still incurs a small loss.



¹Generated by ChatGPT after the prompt: "Can you plot the Hinge loss from -1 to 2?"

Loss function comparisons



Generated by ChatGPT after the prompt: "Can you plot sse, 0/1, and hinge loss together?"



3

³Generated by GPT-4, DALL-E 3 after the prompt: "Hi, could you please make a minimalist logo for a machine learning class..."